

Excel Data Analyst Project Documentation

Sales Data Analysis Using Excel

A Comprehensive Learning Reference for Aspiring Data Analysts

1. Project Overview

The objective of this project is to analyze sales data using Microsoft Excel and generate meaningful business insights through data cleaning, transformation, analysis, and visualization.

This project demonstrates key Excel skills required for a Data Analyst role:

- Data Cleaning & Validation
- Formulas and Functions
- Pivot Tables & Pivot Charts
- Dashboard Creation
- Business Insights Generation

2. Business Problem

Management wants to understand the following business questions to drive data-driven decisions:

- Which products generate the highest revenue?
- Which regions perform best?
- Which salespersons contribute most to sales?
- Product category performance
- Customer purchasing behavior

3. Dataset Information

Dataset Name: Sales Dataset | Records: 2001 Transactions | Source: Online Source

Column Name	Description
Customer_ID	Unique Customer Number
Name	Customer Details
Age	Customer Details
Gender	Customer Details
City	Customer Location
State	Customer Location
Country	Customer Location
Loyalty_Level	Customer Level by Purchase

Column Name	Description
Product_ID	Unique Product Number
Product_Name	Product Details
Category	Product Details
Sub_Category	Product Details
Brand	Product Details
Cost	Product Cost
Stock	Quantity of Product

Column Name	Description
Store_ID	Unique Store Number
Store_Name	Store Details
Region	Store Location
City	Store Location
Store_Type	Store Description

Column Name	Description
Customer_ID	Unique Customer Number
Product_ID	Unique Product Number
Store_ID	Unique Store Number
Quantity	Product Quntity Details
Unit_Price	Product Price
Discount	Price Concession Details
Payment_Type	Payment Mode
Total_Amount	Sales Details

4. Data Cleaning Process

Step	Action
Remove Duplicates	Data → Remove Duplicates
Handle Missing Values	Blank cells identified and corrected
Standardize Data	"north" → "North", "SOUTH" → "South"
Date Formatting	Converted to proper Excel date format (DD-MM-YYYY)
Data Validation	Rules applied for Region, Category, Quantity, Price

5. Excel Functions Used

Formula / Function	Purpose
=[@Cost]*[@Quantity]	Total Spend
=[@Total_Amount]-[@Spend]	Total Profit
=MONTH([@[Order_Date]])	To Extract the Month from Date to do Month wise Analysis
=XLOOKUP()	To Merge the Values From Tables
=IF(ISBLANK([@[Loyalty_Level]]),IFS(AVERAGEIF(\$A:\$A,[@[Customer_ID]],Sales_DB_1[Total_Amount])<=AVERAGEIF(\$H:\$H,\$H\$12,Sales_DB_1[Total_Amount]),\$H\$12,AVERAGEIF(\$A:\$A,[@[Customer_ID]],Sales_DB_1[Total_Amount])<=AVERAGEIF(\$H:\$H,\$H\$6,Sales_DB_1[Total_Amount]),\$H\$6,AVERAGEIF(\$A:\$A,[@[Customer_ID]],Sales_DB_1[Total_Amount])<=AVERAGEIF(\$H:\$H,\$H\$10,Sales_DB_1[Total_Amount]),\$H\$10,AVERAGEIF(\$A:\$A,[@[Customer_ID]],Sales_DB_1[Total_Amount])>=AVERAGEIF(\$H:\$H,\$H\$10,Sales_DB_1[Total_Amount]),\$H\$3),[@[Loyalty_Level]])	For Data Imputation

Formula / Function	Purpose
=IFERROR()	Error Handling
=TEXT / LEFT / RIGHT / MID	Text Manipulation
=DATE / YEAR / MONTH	Date Extraction

6. Pivot Table Analysis

Sales by Region

Region	Revenue
North	647337.6584
South	635213.0957
East	622022.1133
West	270676.406

 **Insight:** North Region generated the highest sales revenue among all four regions.

Sales by Category

Category	Revenue
Sports	541968.2958
Electronics	504311.348
Home	427295.8036
Clothing	379226.2821
Beauty	322447.5439

 **Insight:** Sports is the top-performing category, contributing nearly 50% of total revenue.

Sales By Loyalty_Level

Loyalty_Level	Revenue
Silver	460883.8626
Gold	517281.7332
Bronze	533487.3572

Platinum	663596.8626
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 **Insight:** Platinum is the most Contributing Loyalty Level in Revenue.

7. Visualizations Created

Chart Type	Purpose	Key Insight
Bar Chart	Compare Sales By Loyalty Level	Platinum has highest revenue
Column Chart	Compare Store Sales By Loyalty Level	Electronics leads all categories
Pie Chart	Show Customer Segmentation	Platinum has highest Customer.
Line Chart	Monthly sales trend	Sales peak during festive months
Treemap	Category wise Revenue	Sports Category has highest Revenue.

8. Dashboard Components

A dynamic dashboard was created using the following components:

- KPI Cards — Total Revenue, Total Customer, Total Profit, Total Order, Payment Type
- Pivot Charts — Linked to live data
- Slicers — Region Filter, Category Filter, Payment Type
- Timeline Filter — For date-based filtering

KPI Summary

₹21,70,907 Total Revenue	₹8,95,060 Total Profit	2,001 Total Orders	501 Total Customer
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9. Key Business Insights

- 1. The **sports** category generated the highest revenue.
- 2. **Platinum** customers contributed most of the revenue.
- 3. **Online** stores outperformed physical stores in sales and profit.
- 4. **North region** was the best-performing region, generating the highest sales.

10. Recommendations

Area	Recommendation
Beauty	Expand inventory and run targeted promotions
West Region	Allocate higher marketing budget
Low-Profit Products	Review pricing strategy and supplier costs
Salesperson Training	Use top performers as mentors for the team
Demand Forecasting	Use historical trends for inventory planning

11. Skills Demonstrated

✓ Data Cleaning ✓ Data Validation ✓ Conditional Formatting ✓ Excel Functions & Formulas ✓ Pivot Tables & Pivot Charts	✓ Dashboard Design ✓ KPI Reporting ✓ Data Visualization ✓ Business Insights Generation
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12. Conclusion

This Excel Data Analyst project successfully transformed raw retail sales data into actionable business insights. Using Excel's analytical capabilities — including pivot tables, formulas, dashboards, and visualizations — management can now make informed decisions to improve revenue, profitability, and operational efficiency.